第十四讲
简单静不定梁
合理刚度设计

作业：7-18(b),19(b),22,23,24,30;

第七章　弯曲变形

7.1 挠度与转角

7.2 挠曲轴近似微分方程

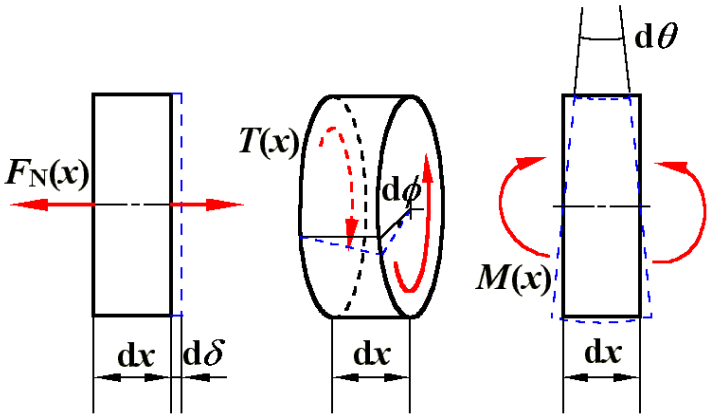
7.3 计算梁位移的积分法

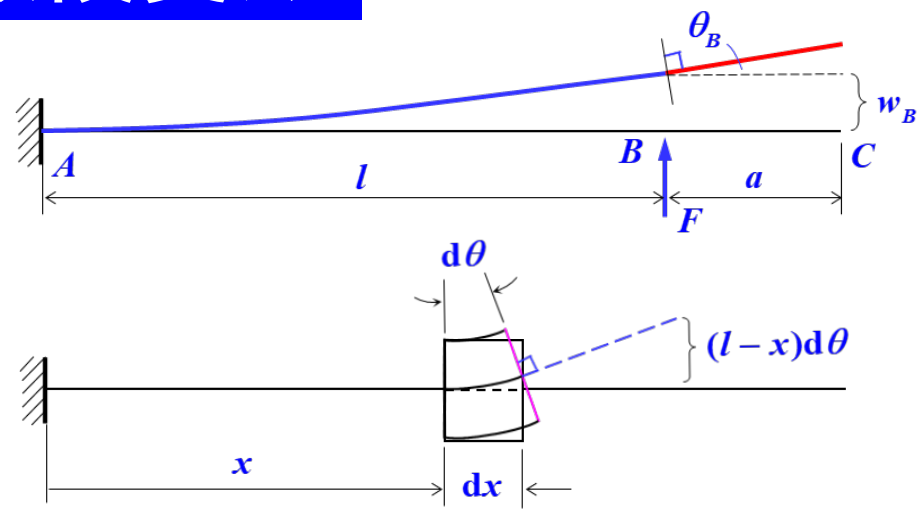
7.4 计算梁位移的叠加法

7.5 简单静不定梁

7.6 梁的刚度条件与合理刚度设计

上讲回顾——认识微段变形





三个平面假设

微段变形

$$d\delta = \frac{F_N}{EA}dx, \quad d\varphi = \frac{T}{GI_P}dx$$

$$d\theta = \frac{dx}{\rho} = \frac{M(x)}{EI_z}dx$$

应力

$$\Delta l = \int d\delta$$

$$\Delta\varphi = \int d\varphi$$

梁变形？

揭示了拉压弯扭的基本特征

$$d\theta_B = d\theta, \quad dw_B = (l-x)d\theta$$

影响系数——微段变形的贡献

$$\mu_{\theta B} = 1, \quad \mu_{wB} = l - x$$

$$\theta_B = \int_0^l \mu_{\theta B}\, d\theta, \quad w_B = \int_0^l \mu_{wB} d\theta$$

$$\theta_C = \theta_B, \quad w_C = w_B + \theta_B a$$

微段变形累加形成截面挠度与转角

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第七章 弯曲变形

7.5 简单静不定梁

1、多余约束与相当系统

2、简单静不定梁分析方法

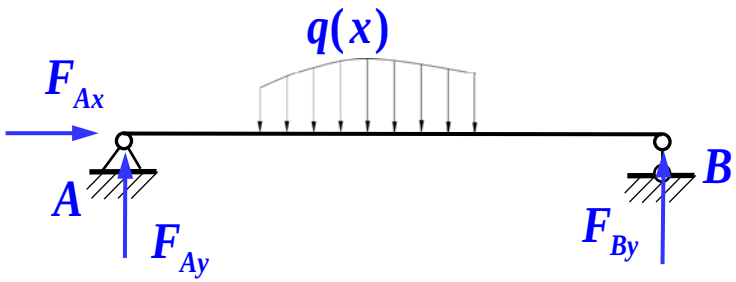
3、例题

4、对称与反对称问题

1、多余约束与相当系统

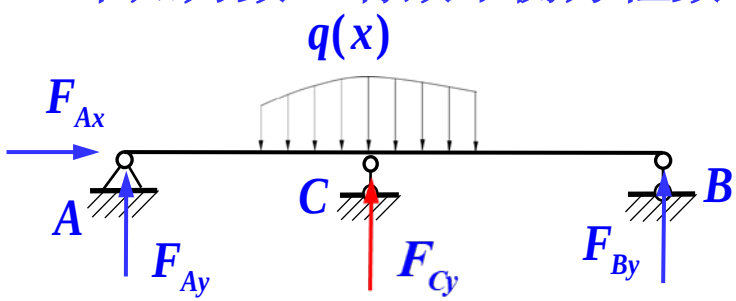
静定梁

未知力数＝有效平衡方程数

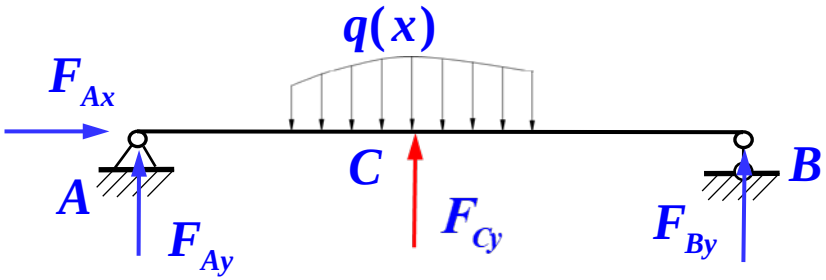


静不定梁

未知力数 > 有效平衡方程数



相当系统



- 多余约束的作用由多余约束力代替；
- 与静不定梁受力相当的静定梁；
- 去掉的多余约束处，满足变形协调条件。

$w_C = 0$ 变形协调条件

多余约束力 ← 多余约束

维持平衡不需要

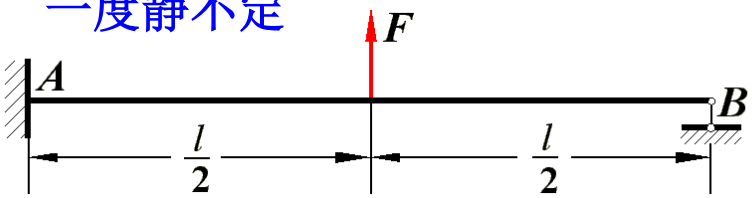
与静定问题的根本区别

求解的关键

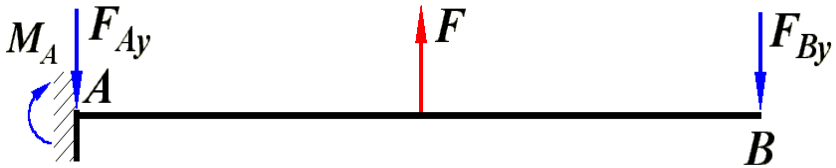
2、简单静不定梁分析方法

图示梁，EI为常数，求支反力

一度静不定



选 B支座为多余约束



相当系统

变形协调条件一 $w_B = 0$

$$w_B = \frac{F\left(l/2\right)^3}{3EI} + \frac{F\left(l/2\right)^2}{2EI}\cdot\frac{l}{2} - \frac{F_{By}l^3}{3EI} \Rightarrow F_{By} = \frac{5F}{16}$$

静不定问题得解

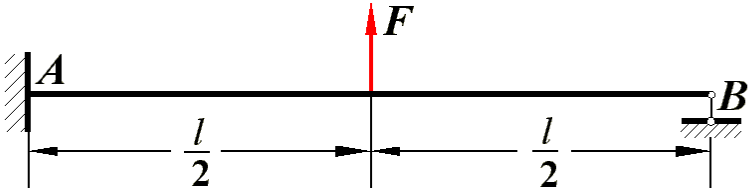
- 求支反力
- 求内力、应力
- 求变形

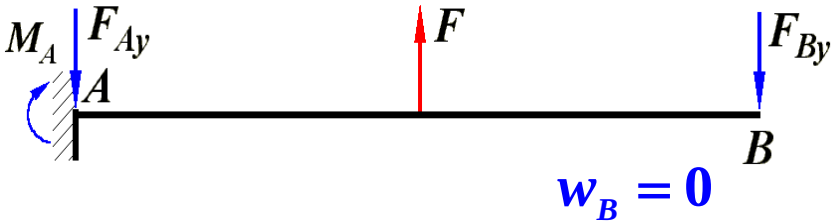
} 与静定问题同样分析

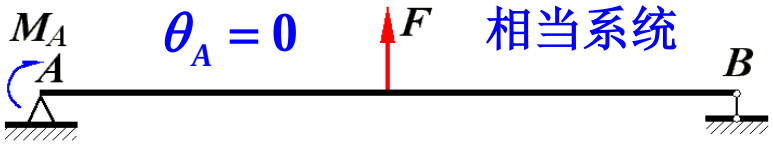
2、简单静不定梁分析方法

分析方法与步骤

- 判断静不定度；
- 选择多余约束，建立相当系统，给变形协调条件；
- 计算相当系统在多余约束处的位移（挠度、转角）；
- 代入协调条件，得到多余约束力；
- 之后分析与静定问题相同。







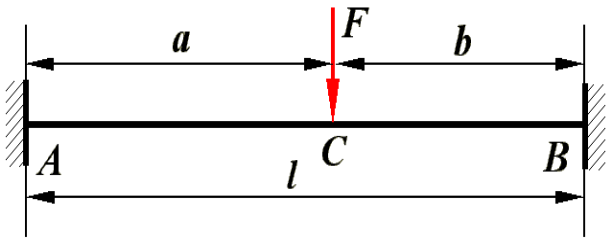
多余约束、相当系统不唯一

3、例题

例7-8 图示梁，EI为常数，求支反力。

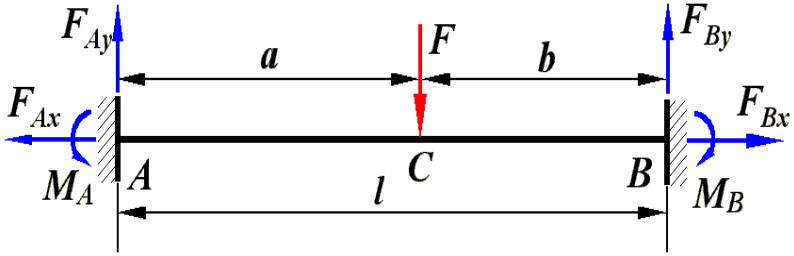
解： 1）问题分析 三度静不定

2）相比挠度 不计轴向位移，略去轴向反力 两个未知约束力



3）建立相当系统

变形协调条件 $\theta_A = 0,\ \theta_B = 0$

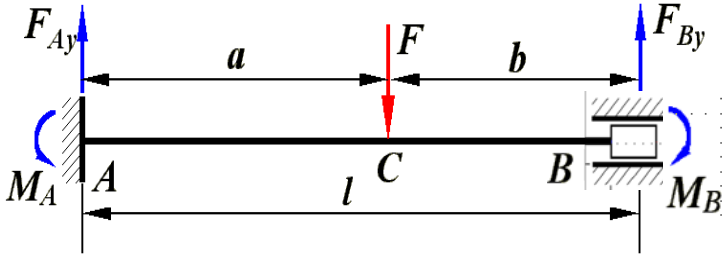


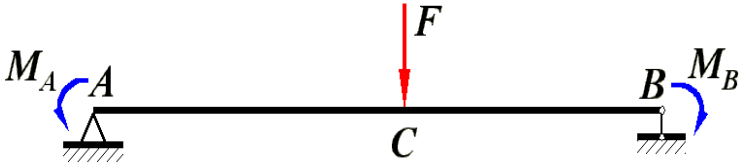
$$\theta_A = \theta_{A,F} + \theta_{A,M_A} + \theta_{A,M_B} = -\frac{Fab(l+b)}{6EIl} + \frac{M_A l}{3EI} + \frac{M_B l}{6EI} = 0$$

$$\theta_B = \theta_{B,F} + \theta_{B,M_A} + \theta_{B,M_B} = \frac{Fab(l+a)}{6EIl} - \frac{M_A l}{6EI} - \frac{M_B l}{3EI} = 0$$

$$M_A = \frac{Fab}{l^2}b,\quad F_{Ay} = F\left(\frac{b^2}{l^2} + \frac{2ab}{l^3}b\right)$$

$$M_B = \frac{Fab}{l^2}a,\quad F_{By} = F\left(\frac{a^2}{l^2} + \frac{2ab}{l^3}a\right)$$





- 内力按刚度分配；
- 若载荷居中，利用对称性可简化分析

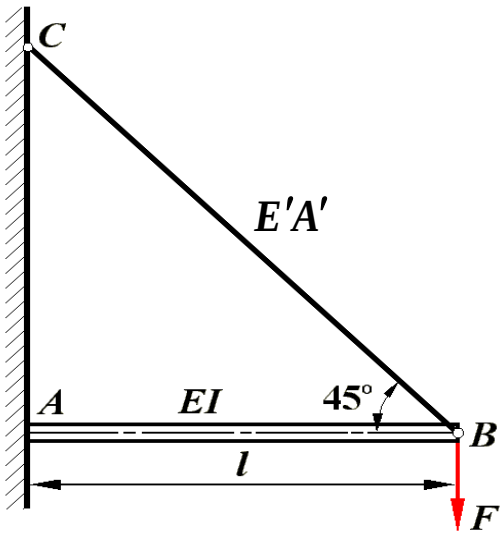
3、例题

例7-9 图示梁杆结构，求斜杆轴力。

解：1）问题分析

钢丝BC提供了**弹性约束**，一度静不定；

相比挠度，梁的轴向位移、轴向压缩均可略去。



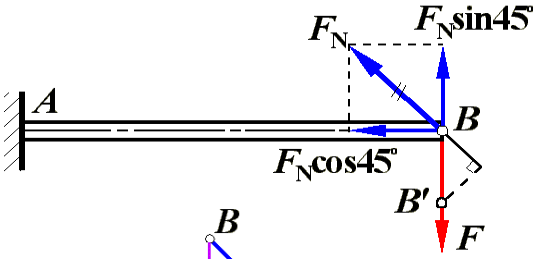
2）建立相当系统，画变形图

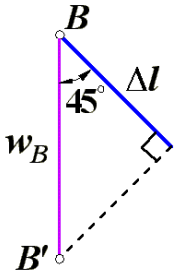
变形协调条件 $w_B = \sqrt{2}\Delta l$

$$w_B = \frac{\left(F - F_N/\sqrt{2}\right)l^3}{3EI}$$

$$\Delta l = \frac{F_N \cdot \sqrt{2}l}{E'A'}$$

$$\Rightarrow \quad F_N = \frac{\sqrt{2}F}{1 + 6\dfrac{EI}{E'A'l^2}}$$





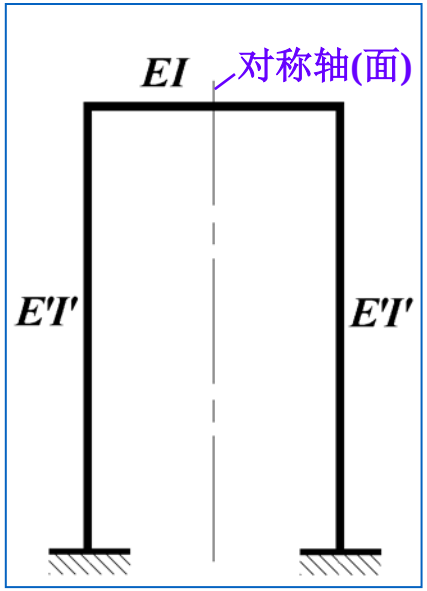
讨论

1）若梁杆材料相同、截面尺寸相当，$F_N = \sqrt{2}F$

2）细杆吊粗梁，梁杆分担。

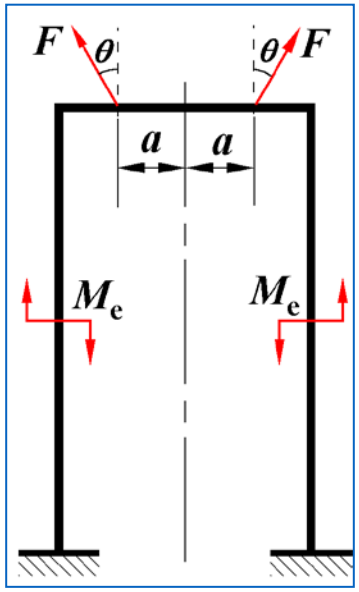
4、对称与反对称问题

对称结构



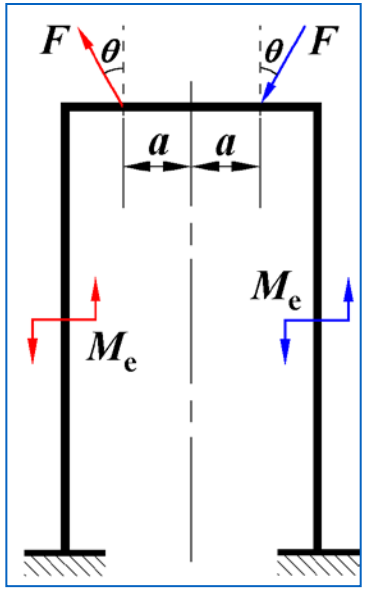
结构具有对称的形状、尺寸、力学性能与约束条件

对称载荷



载荷作用点(或面)、大小、方位与指向(或转向)均对称

反对称载荷



载荷作用点(或面)、大小与方位均对称，但指向(或转向)反对称

4、对称与反对称问题

对称定律

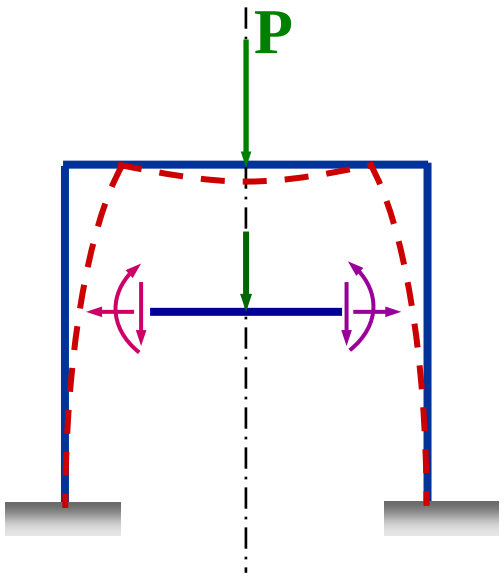
对称结构

对称载荷 → 内力、变形对称

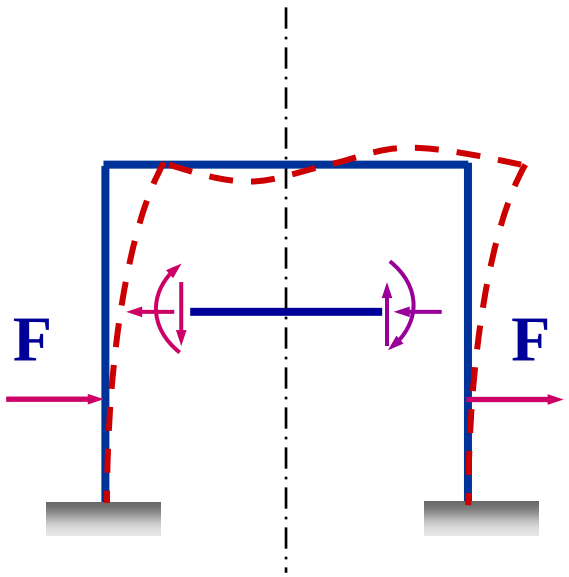
反对称载荷 → 内力、变形反对称

对称的截面上

对称承载



反对称承载

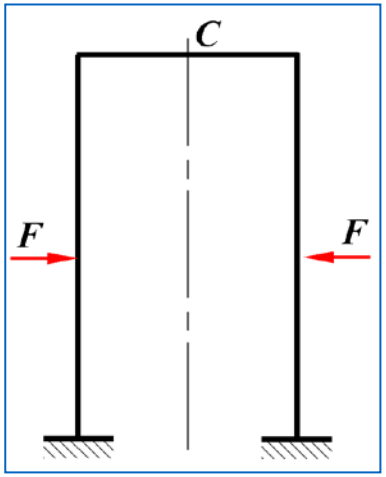


4、对称与反对称问题

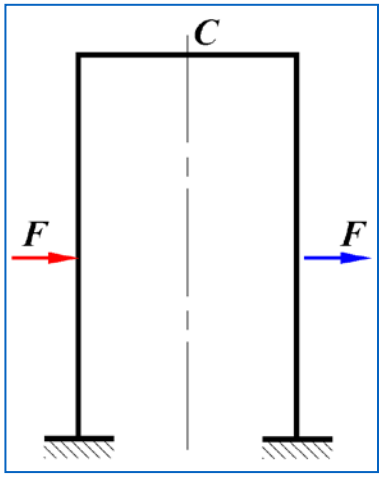
对称面上受力与变形特点

考察对称面两侧截面的内力与位移

对称承载



反对称承载



内力
- 对称或反对称条件
- 互为作用力与反作用力

对称定律

牛顿第三定律

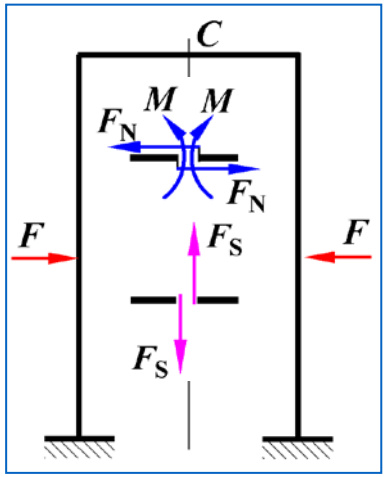
变形
- 对称或反对称条件
- 变形连续条件

4、对称与反对称问题

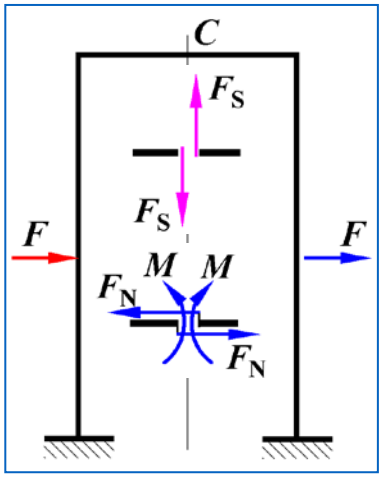
对称面上受力与变形特点

考察对称面两侧截面的内力与位移

对称承载



反对称承载



仅存在对称性内力—弯矩M与轴力F_N，
而剪力 $F_S = 0$
轴向位移 $\Delta = 0$，转角 $\theta = 0$

仅存在反对称性内力—剪力 F_S
而$M=0$，$F_N = 0$
横向挠度 $f = 0$

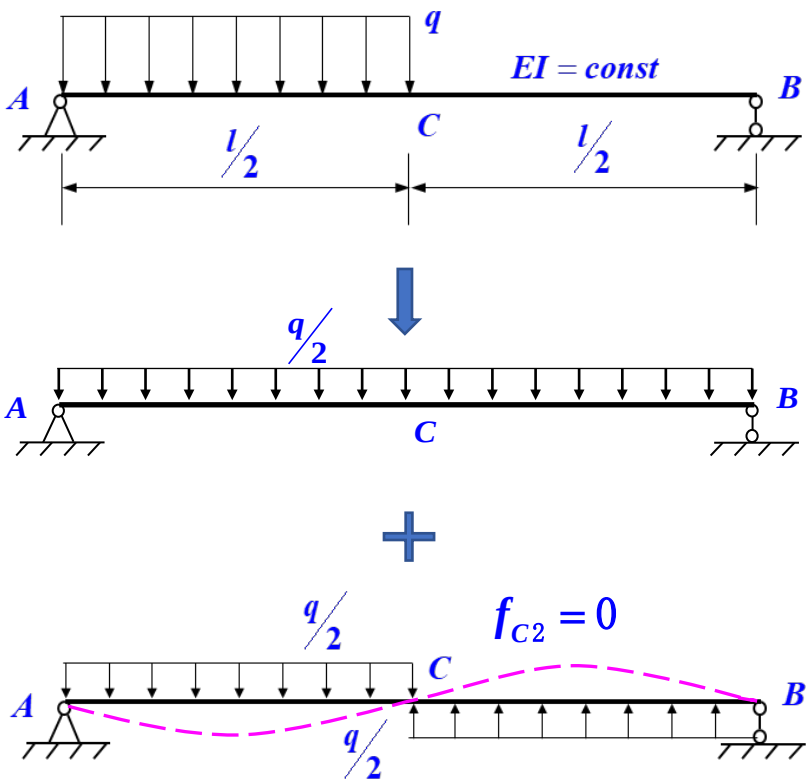
说明：利用问题的对称性，可直接确定对称面上的多余约束力，从而简化计算分析，但并不降低静不定度

4、对称与反对称问题

例7-7 求C截面挠度和转角

解：

载荷的对称与反对称分解



$$\theta_{C1}=0,\quad f_{C1}=\frac{5\left(\frac{q}{2}\right)l^4}{384EI}(\downarrow)$$

$$\theta_{C2}=\frac{\frac{q}{2}\left(\frac{l}{2}\right)^3}{24EI}(\circlearrowleft)$$

$$f_C=f_{C1},\quad \theta_C=\theta_{C2}$$

第七章 弯曲变形

7.6 梁的刚度条件与合理刚度设计

1、梁的刚度条件

2、梁的合理刚度设计

3、例题

1、梁的刚度条件

最大位移控制

$|w|_{\max} \le [\delta]$　　$[\delta]$—许用挠度

$|\theta|_{\max} \le [\theta]$　　$[\theta]$—许用转角

桥式起重机梁：$[\delta] = \dfrac{l}{750} \sim \dfrac{l}{500}$

一般用途轴：$[\delta] = \dfrac{3l}{10000} \sim \dfrac{5l}{10000}$

指定截面位移控制

$|w| \le [\delta]$　　$|\theta| \le [\theta]$

例如滑动轴承处

$[\theta] = 0.001\,\text{rad}$

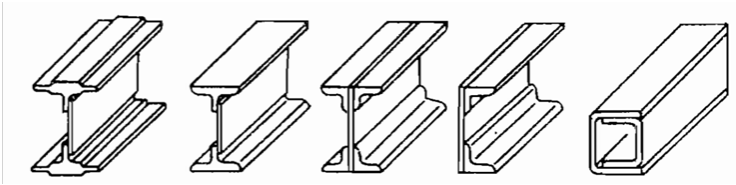
2、梁的合理刚度设计

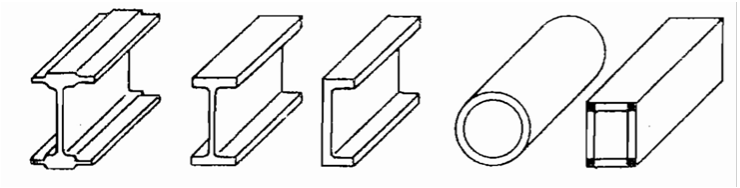
$$\frac{d^2w}{dx^2}=\frac{M(x)}{EI}$$

截面惯性矩大

工字形、盒式、空心圆、T形、槽形、Z字形、L形等薄壁截面。

组合式翼梁





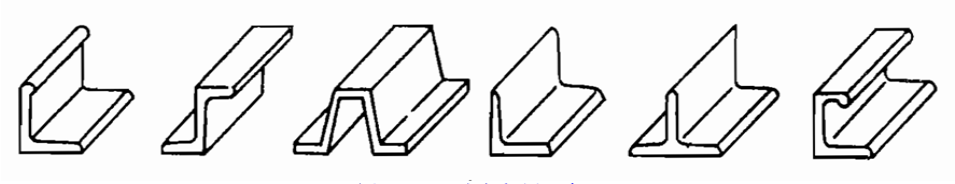
整体式翼梁

材料弹性模量高

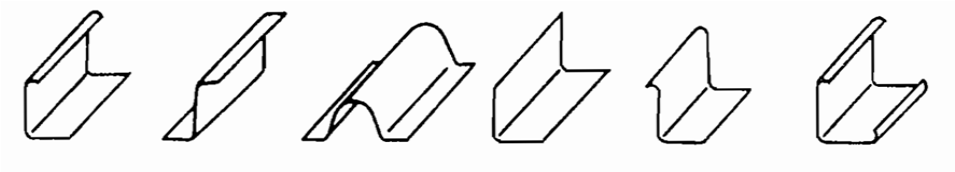
各种牌号的钢材（或铝合金），弹性模量基本相同。

钢与合金钢：200~220 *GPa*;

铝合金：70~72 *GPa*.



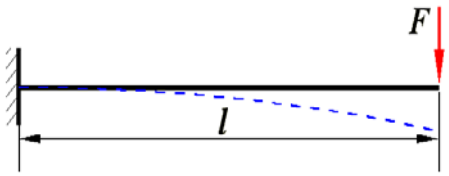
挤压型材桁条

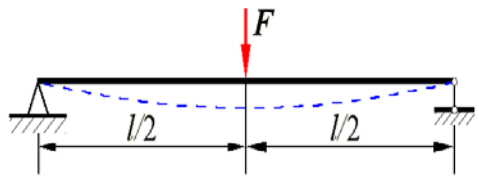


钣弯型材桁条

2、梁的合理刚度设计

合理选取梁的跨度





$$|M|_{\max} = Fl \qquad \delta_{\max} = \frac{Fl^3}{3EI}$$

$$M_{\max} = \frac{Fl}{4} \qquad \delta_{\max} = \frac{Fl^3}{48EI}$$

$$|M|_{\max} \propto l \qquad \delta_{\max} \propto l^3$$

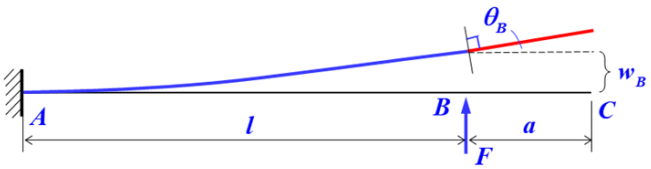
$\dfrac{l^3}{EI}$ ~柔度系数，$\dfrac{EI}{l^3}$ ~刚度系数

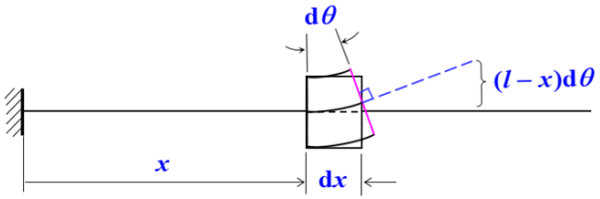
例如 l 缩短 20%，$\delta_{\max}$ 将减少 48.8%

跨度微小改变，将导致挠度显著改变

局部加强

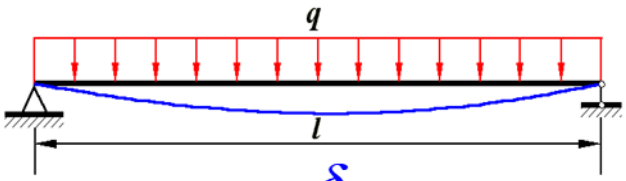
显著提高局部强度，
对整体刚度无大影响



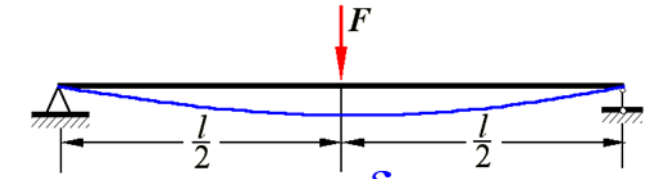


2、梁的合理刚度设计

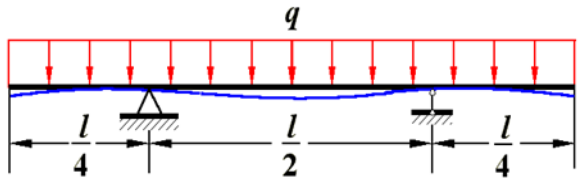
合理安排约束与加载方式



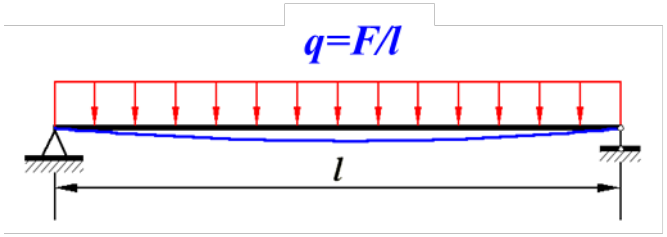
$\delta_{1,\max}$



$\delta_{1,\max}$



$\delta_{2,\max}$



$\delta_{2,\max}$

$$\frac{\delta_{2,\max}}{\delta_{1,\max}} = 8.75\ \%$$

$$\frac{\delta_{2,\max}}{\delta_{1,\max}} = 12.5\ \%$$

增加约束，制成静不定梁

本讲：简单静不定梁

解除多余约束

下讲：应力状态分析